

11/06/26 "DAY 16" "Kamineni Dhana Sri Keerthi"

Defect free TSV Equations. [RLAC]

Equation 01:

$$R_{TSV} = \sqrt{[R_{dc,TSV}]^2 + [R_{ac,TSV}]^2} \quad [\Omega]$$

where

$$R_{dc,TSV} = \left[\frac{\rho_{TSV} h_{TSV}}{\pi (r_{TSV})^2} \right]$$

$$R_{ac,TSV} = \left[\frac{\rho_{TSV} h_{TSV}}{[2\pi r_{TSV} \delta_{SD,TSV} - \pi \delta_{SD,TSV}^2]} \right] \quad [\Omega]$$

$$\delta_{SD,TSV} = \frac{1}{\sqrt{\pi f \mu_{TSV} \sigma_{TSV}}} \quad [m]$$

$$\mu_{TSV} = \mu_0 \mu_{r,TSV}$$

Equation 02:

$$L_{TSV} = \left[\frac{1}{2} \left[\frac{\mu_0 \mu_{r,TSV} h_{TSV}}{2\pi} \ln \left[\frac{\rho_{TSV}}{r_{TSV}} \right] \right] \right] \quad [H]$$

Equation 03:

$$G_{Si} = \left[\frac{\pi \sigma_{Si} h_{TSV}}{\cosh^{-1} \left[\frac{\rho_{TSV}}{d_{TSV}} \right]} \right] \quad [S]$$

Equation 04:

$$C_{ox} = \frac{2\pi \epsilon_0 \epsilon_{r,ox} \left[\frac{h_{TSV}}{r_{TSV}} \right]}{\ln \left[\frac{r_{TSV} + t_{ox}}{r_{TSV}} \right]} \quad [F]$$

Equation 05:

$$C_{dep} = \frac{2\pi \epsilon_0 \epsilon_{r,si} \left[\frac{h_{TSV}}{r_{TSV}} \right]}{\ln \left[\frac{r_{TSV} + t_{ox} + t_{dep}}{r_{TSV} + t_{ox}} \right]} \quad [F]$$

Equation 06:

$$C_{si} = \frac{\pi \epsilon_0 \epsilon_{r,si} h_{TSV}}{\cosh^{-1} \left[\frac{r_{TSV}}{d_{TSV}} \right]} \quad [F]$$

Equation 07:

$$C_{ins} = \frac{C_{dep} C_{ox}}{C_{dep} + C_{ox}} \quad [F]$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$$

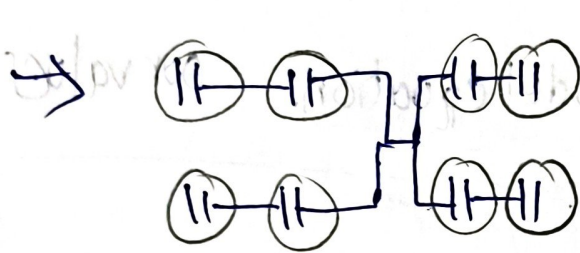
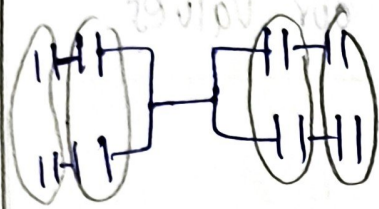
$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

value table.

Symbol	value	Symbol	value	Symbol	value
r_{TSV}	10 μm	L_{sub}	3.06 μm	F	100 kHz
d_{TSV}	4.94 μm	σ_{si}	10 S/m	μ_{TSV}	1
r_{TSV}	2.47 μm	σ_{TSV}	5.9e7 S/m	d_{si}	0
h_{TSV}	5 μm	ρ_{TSV}	1.68e-8 Ωm	d_{ox}	0
t_{ox}	0.13 μm	$\epsilon_{r,si}$	11.9	σ_{SiO_2}	10 ⁻¹⁶ S/m
t_{dep}	0.87 μm	$\epsilon_{r,ox}$	4		

Editing (Defect free).

▲ For matlab code. $\rightarrow$ in C_{ox} and C_{dep} we will take h_{TSV} instead of $\frac{h_{TSV}}{2}$ in while doing hspice we will divide it as separate



▲ Since $\frac{P_{TSV}}{d_{TSV}} < 10 \mu m$ we will use $\cosh^{-1}$ instead of $\ln$
Reference from High frequency TSV [scalable model] paper.

IRDS 2022 Executive summary TSV diagram

2021 IRDS metrology table MFT-4.

~~*~~ Values from this Reference.

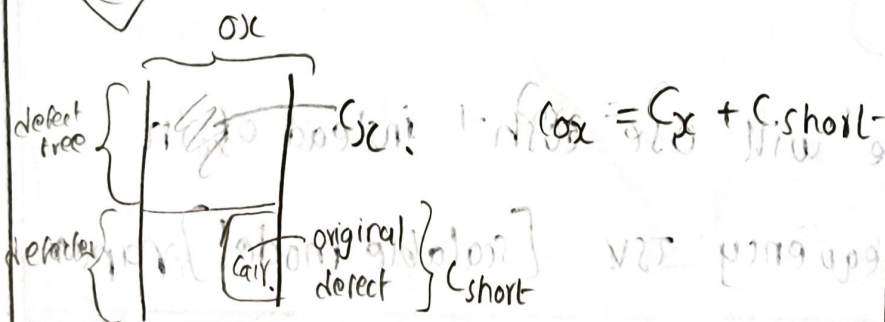
Editing [Defected CG]

01 $RLCG \rightarrow$ cliu Model (their eqⁿ ^{our} their values)

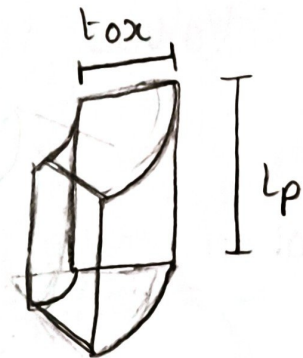
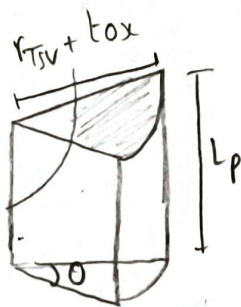
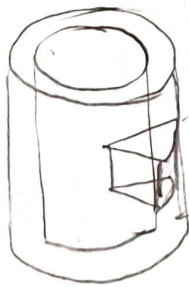
$\left. \begin{matrix} C_R \\ C_L \end{matrix} \right\}$ defect free $\left. \begin{matrix} C \\ G \end{matrix} \right\}$ cliu equation $\rightarrow$ our values

02 $RLCG \rightarrow$ our proposed model

$\left. \begin{matrix} R \\ L \end{matrix} \right\}$ defect free $\left. \begin{matrix} C \\ G \end{matrix} \right\}$ our model equation $\rightarrow$ our values



Editing [Defected RL]



$$L_{shf} = \frac{\mu}{2p\theta} \ln \left(\frac{r_{TSV} + tox}{r_{TSV}} \right)$$

$$L_{shf} = \frac{\mu \mu_{TSV} tox}{2\pi} \left[\cosh^{-1} \left(\frac{tox}{\sqrt{w_{eq}^2 + t_{ox}^2}} \right) - \sqrt{2t_{ox}^2 + w_{eq}^2} + \frac{tox}{4} \right. \\ \left. + \sqrt{w_{eq}^2 + t_{ox}^2} \right]$$

$$w_{eq} = \theta \left(r_{TSV} + \frac{tox}{2} \right)$$

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Defected TSV $[C_{leak} \& G_{leak}] \rightarrow RLC$ with $C_{leak} \& G_{leak}$

Equation 01:

$$R_{TSV} = \sqrt{[R_{dc,TSV}]^2 + [R_{ac,TSV}]^2}$$

where

$$\rightarrow R_{dc,TSV} = \frac{\rho_{TSV} h_{TSV}}{\pi [r_{TSV}]^2} [\Omega]$$

$$\rightarrow R_{ac,TSV} = \frac{\rho_{TSV} h_{TSV}}{2\pi r_{TSV} \delta_{SD,TSV} - \pi \delta_{SD,TSV}^2} [\Omega]$$

$$\rightarrow \delta_{SD,TSV} = \frac{1}{\sqrt{\pi f \mu_{TSV} \sigma_{TSV}}} [m]$$

Equation 02:

$$L_{TSV} = \frac{1}{2} \left[\frac{\mu_0 \mu_{r,TSV} h_{TSV}}{2\pi} \right] \ln \left[\frac{\rho_{TSV}}{r_{TSV}} \right] [H]$$

Equation 03:

$$G_{Si} = \frac{\pi \sigma_{Si} [h_{TSV} - l_p]}{\cosh^{-1} \left[\frac{\rho_{TSV}}{d_{TSV}} \right]} [S]$$

Equation 04:

$$C_{ox} = \frac{2\pi \epsilon_0 \epsilon_{r,ox} [h_{TSV}]}{\ln \left[\frac{r_{TSV} + t_{ox}}{r_{TSV}} \right]} [F]$$

Equation 05:

$$C_{dep} = \frac{2\pi \epsilon_0 \epsilon_r, si [h_{TSV}]}{\ln \left[\frac{r_{TSV} + t_{ox} + t_{dep}}{r_{TSV} + t_{ox}} \right]} \quad [F]$$

Equation 06:

$$C_{si} = \frac{\pi \epsilon_0 \epsilon_r, si [h_{TSV} - L_p]}{\cosh^{-1} \left[\frac{r_{TSV}}{d_{TSV}} \right]} \quad [F]$$

Equation 07:

$$C_{ins} = \frac{C_{dep} C_{ox}}{C_{dep} + C_{ox}} \quad [F]$$

Model 01: "Chao Liu" C_{leak} and G_{leak} .

Equation 08:

$$C_{leak} = \frac{2\pi \epsilon_0 \epsilon_r, ox [h_{TSV}] + 2\pi \epsilon_0 \epsilon_r, air J r_{TSV}}{\ln \left[\frac{r_{TSV} + t_{ox}}{r_{TSV} + t_{ox}} \right]} \quad [F]$$

where $J = \frac{0.1 p}{2\pi r_{TSV}}$

$\epsilon_{ox} = \frac{6.5 \times 10^{-14}}{0.12}$

Equation 09:

$$G_{leak} = \frac{2\pi \sigma_{SiO_2} h_{TSV} - 4\pi^2 J f \epsilon_0 \epsilon_r, air r_{TSV}}{\ln \left[\frac{r_{TSV} + t_{ox}}{r_{TSV}} \right]}$$

Model 02: Proposed Model C_{leak} and G_{leak} .

Equation 10:

$$C_{short} = C_{SiO_2, remain} + C_{air}$$

$$\rightarrow C_{air} = \frac{\theta \epsilon_0 \epsilon_{r, air} L_p}{\ln \left[\frac{r_{TSV} + t_{ox}}{r_{TSV}} \right]} \quad [F]$$

$$\rightarrow C_{SiO_2, remain} = \frac{[2\pi - \theta] \epsilon_0 \epsilon_{r, ox} L_p}{\ln \left[\frac{r_{TSV} + t_{ox}}{r_{TSV}} \right]} \quad [F]$$

$$C_{short} = \frac{[2\pi - \theta] \epsilon_0 \epsilon_{r, ox} L_p + \theta \epsilon_0 \epsilon_{r, air} L_p}{\ln \left[\frac{r_{TSV} + t_{ox}}{r_{TSV}} \right]} \quad [F]$$

$$C_x = \frac{2\pi \epsilon_0 \epsilon_{r, SiO_2} (h_{TSV} - L_p)}{\ln \left(\frac{r_{TSV} + t_{ox}}{r_{TSV}} \right)}$$

Equation 11

$$C_{ox} = C_x + C_{short}$$

$$\rightarrow C_x = \frac{2\pi \epsilon_0 \epsilon_{r, ox} [h_{TSV} - L_p]}{\ln \left[\frac{r_{TSV} + t_{ox}}{r_{TSV}} \right]} \quad [F]$$

$$\rightarrow C_{air} = \frac{\theta \epsilon_0 \epsilon_{r, air} L_p}{\ln \left[\frac{r_{TSV} + t_{ox}}{r_{TSV}} \right]} \quad [F]$$

Equation 12

$$G_{short} = G_{SiO_2, remain} + G_{air}$$

$$G_{air} = \frac{\theta \epsilon_0 \epsilon_{r, air} L_p f}{\ln \left[\frac{r_{TSV} + t_{ox}}{r_{TSV}} \right]} \quad [S]$$

$$G_{ox} = G_{short}$$

$$G_{\text{SiO}_2, \text{remain}} = \frac{[2\pi - \theta] \sigma_{\text{SiO}_2} L_p}{\ln \left[\frac{r_{\text{TSV}} + t_{\text{ox}}}{r_{\text{TSV}}} \right]} \quad [5]$$

$$G_{\text{short}} = \frac{2\pi - \theta \sigma_{\text{SiO}_2} L_p + \theta \epsilon_0 \epsilon_{\text{ratr}} L_p f}{\ln \left[\frac{r_{\text{TSV}} + t_{\text{ox}}}{r_{\text{TSV}}} \right]} \quad [5]$$

$$G_{\text{ox}} = G_{\text{short}}.$$

Symbol	Value.	Symbol	Value.	
ρ_{TSV}	10 μm	$\epsilon_{\text{r, Si}}$	11.9	
d_{TSV}	4.94 μm	$\epsilon_{\text{r, ox}}$	4	
r_{TSV}	2.47 μm	f	100 GHz	
h_{TSV}	15 μm	$\mu_{\text{r, TSV}}$	1	
t_{ox}	0.13 μm	δ_{Si}	0	$\text{SiO}_2 = \text{ox}$
t_{dep}	0.87 μm	δ_{ox}	0	
σ_{Si}	10 S/m	θ	$\pi/2$	
σ_{TSV}	5.9e7 S/m	L_p	4 μm	
σ_{SiO_2}	10^{-16} S/m	J	40%	
ρ_{TSV}	1.68e-8 Ωm			